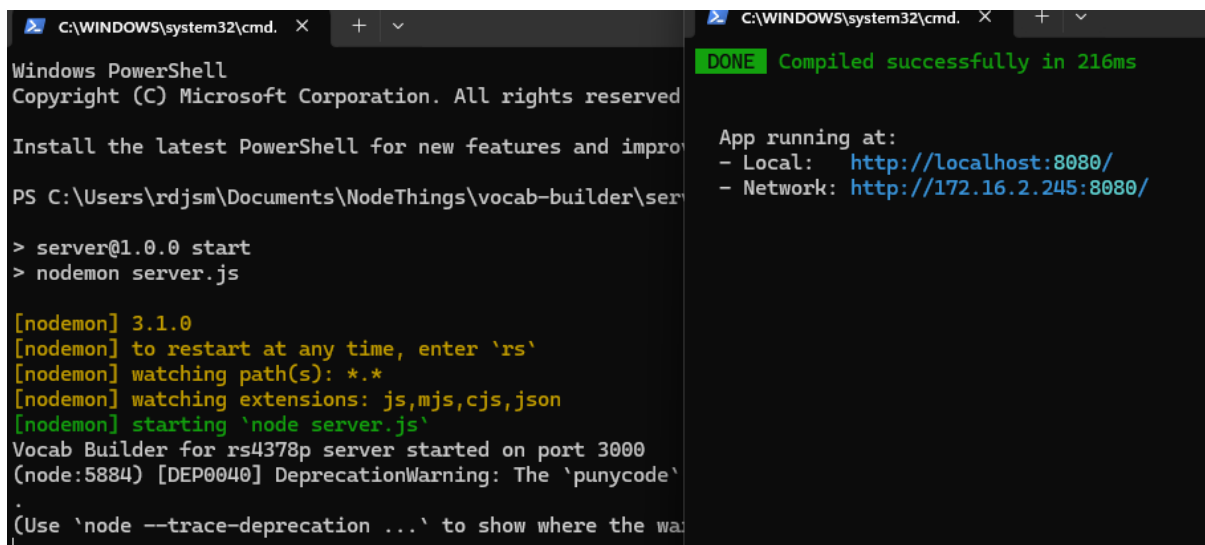


## CRUD System Evidence



```
C:\WINDOWS\system32\cmd. x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

PS C:\Users\rdj\Documents\NodeThings\vocab-builder\server> server@1.0.0 start
> nodemon server.js

[nodemon] 3.1.0
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Vocab Builder for rs4378p server started on port 3000
(node:5884) [DEP0040] DeprecationWarning: The `punycode` module is deprecated. Please use `node:punycode` instead.
(Use `node --trace-deprecation ...` to show where the warning was created)

C:\WINDOWS\system32\cmd. x + v
DONE Compiled successfully in 216ms

App running at:
- Local: http://localhost:8080/
- Network: http://172.16.2.245:8080/
```

Figure 24 - Back-end and front-end servers starting

Figure 24 shows both front-end and back-end servers being started through triggering NPM scripts for each.

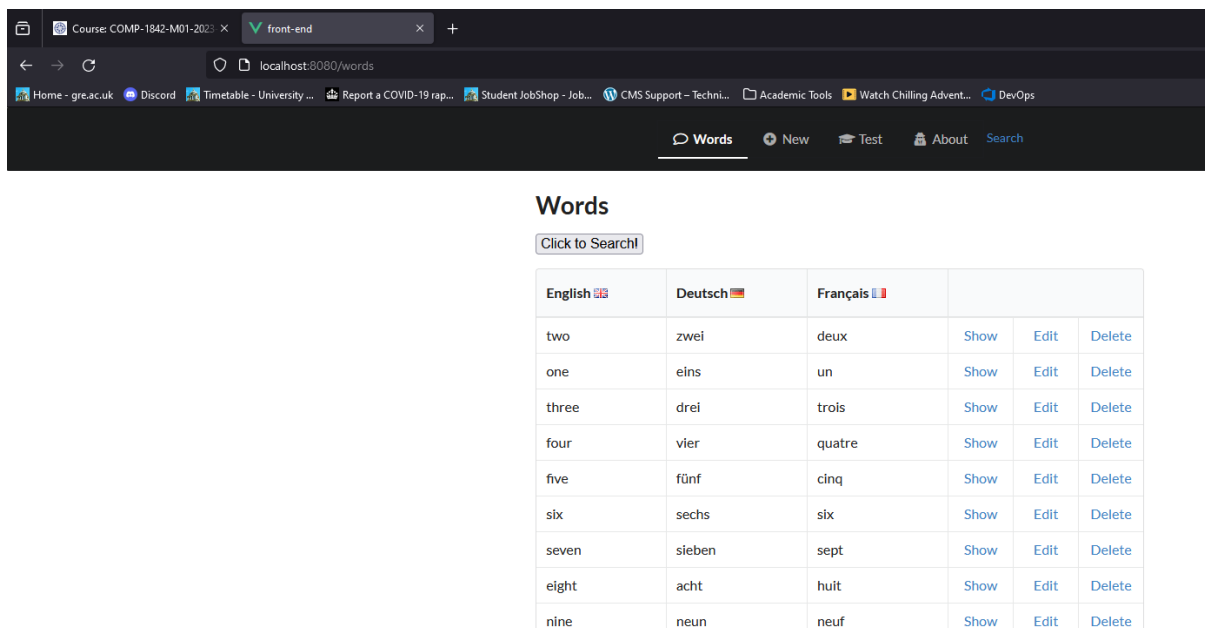
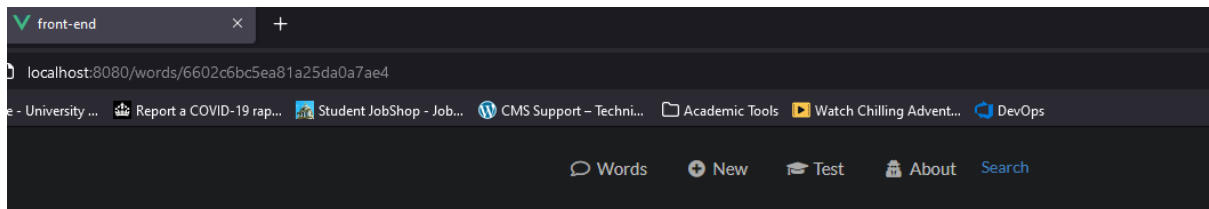


Figure 25 - Default home page

Figure 25 depicts the default view upon navigating to localhost on port 8080 with the servers running. This is a list of all current words with three languages, each can be viewed independently, edited or deleted as required.



## Show Word

|         |      |
|---------|------|
| German  | zwei |
| English | two  |
| French  | deux |

[Edit word](#)

Figure 26 - Viewing a single word

Figure 26 is the display of a single word, and the URL shown includes the unique identifier for the word used in the database.

## Edit Words

|         |         |
|---------|---------|
| German  | zwei    |
| English | twooooo |
| French  | deux    |

[Submit](#)

Figure 27 - View for editing a word

## Words

[Click to Search!](#)

| English | Deutsch | Français |      |      |        |
|---------|---------|----------|------|------|--------|
| twooooo | zwei    | deux     | Show | Edit | Delete |
| one     | eins    | un       | Show | Edit | Delete |
| three   | drei    | trois    | Show | Edit | Delete |

Figure 28 - Result of an edited word

Figures 27 and 28 shows the word “two” being changed to “twooooo” and the result in the default word list with the change made.

Words

New

Test

About

Search

New Word Creation

German

test1

English

test2

French

test3

Submit

Figure 29 - View for creating a new word

|          |          |          |      |      |        |
|----------|----------|----------|------|------|--------|
| fourteen | vierzehn | quatorze | Show | Edit | Delete |
| fifteen  | fünfzehn | quinze   | Show | Edit | Delete |
| test2    | test1    | test3    | Show | Edit | Delete |

Figure 30 - Result of adding new word

Figures 29 and 30 represent what happens when a new word is added, for all three languages, and is then added to the main list at the end. The only issue with this system is that to re-order words currently you need to use the edit function and replace them. For example to swap fourteen and fifteen you would edit fourteen, change it to fifteen then edit the original fifteen and swap it to fourteen.

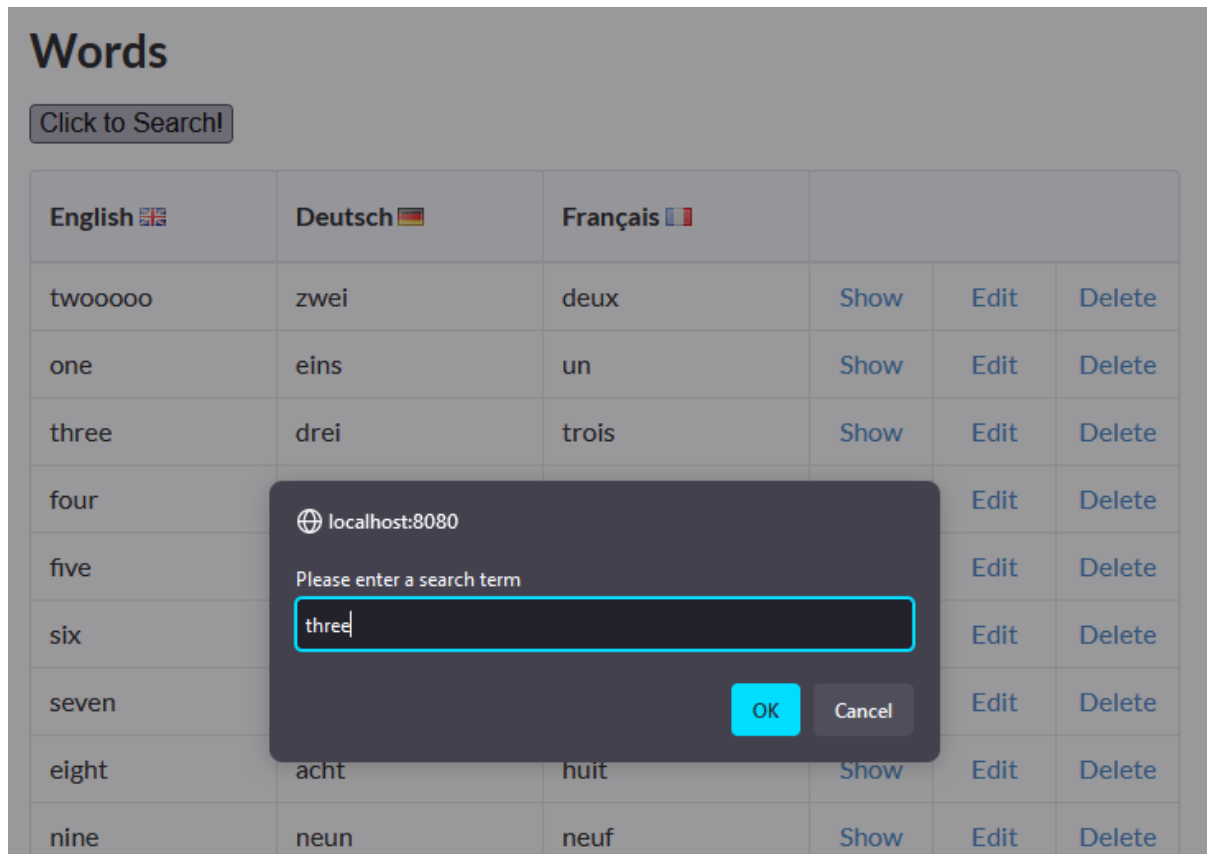


Figure 31 - Search box for a term

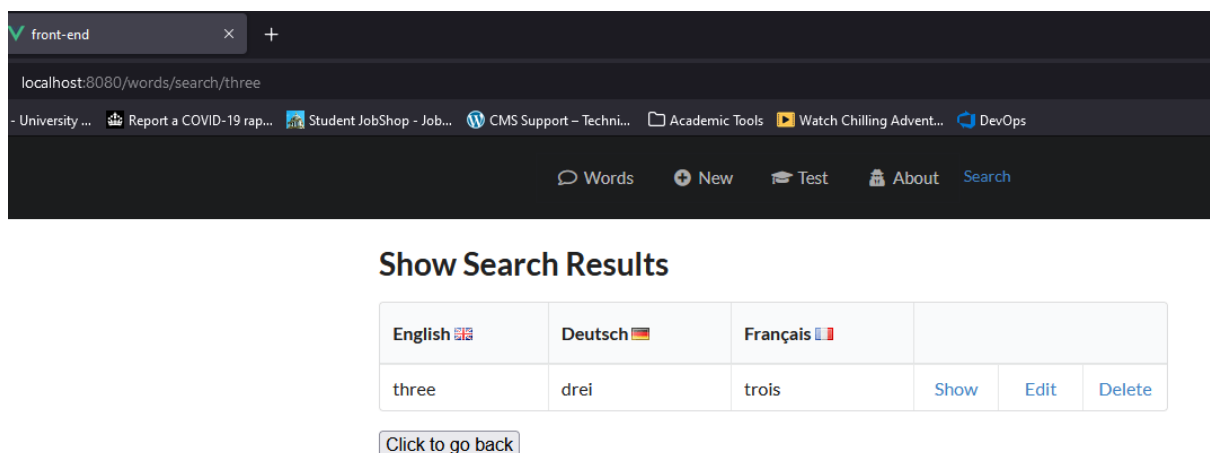


Figure 32 - Search results view

Figures 31 and 32 highlight the search functionality of the system, clicking the search button displays a popup window alert box, whatever term inserted there is then searched for in each language. If a suitable result is found it redirects the user to view this word.

The search is based on the URL, so it can be bookmarked for ease of access at a later time or shared with others if it were a live internet based system.

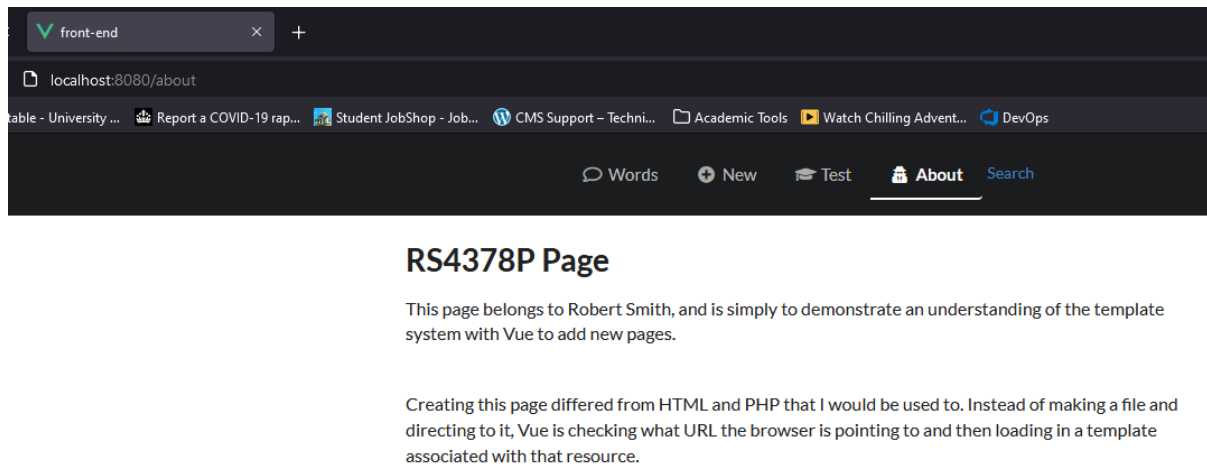


Figure 33 - About Me page view

Figure 33 shows the about section, a simple page created within the Vue template functionality.

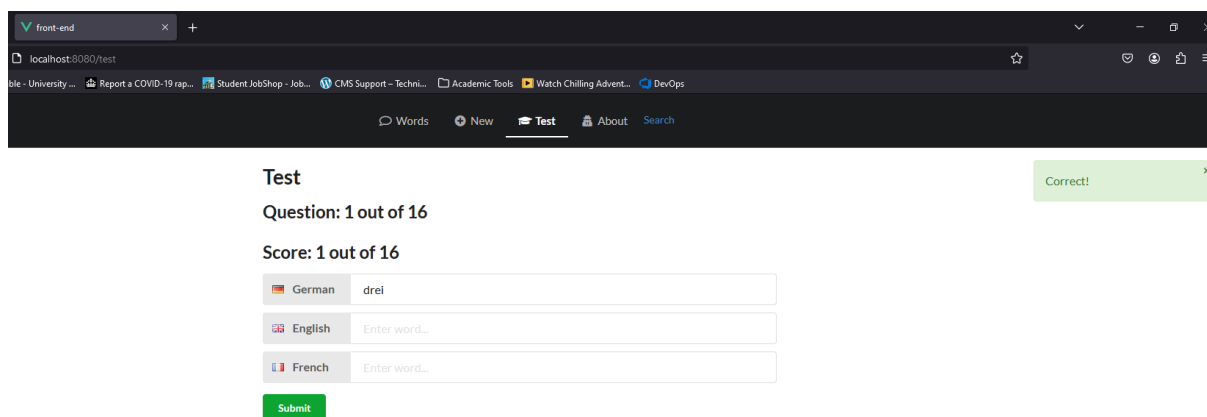


Figure 34 - Test view of correct answer

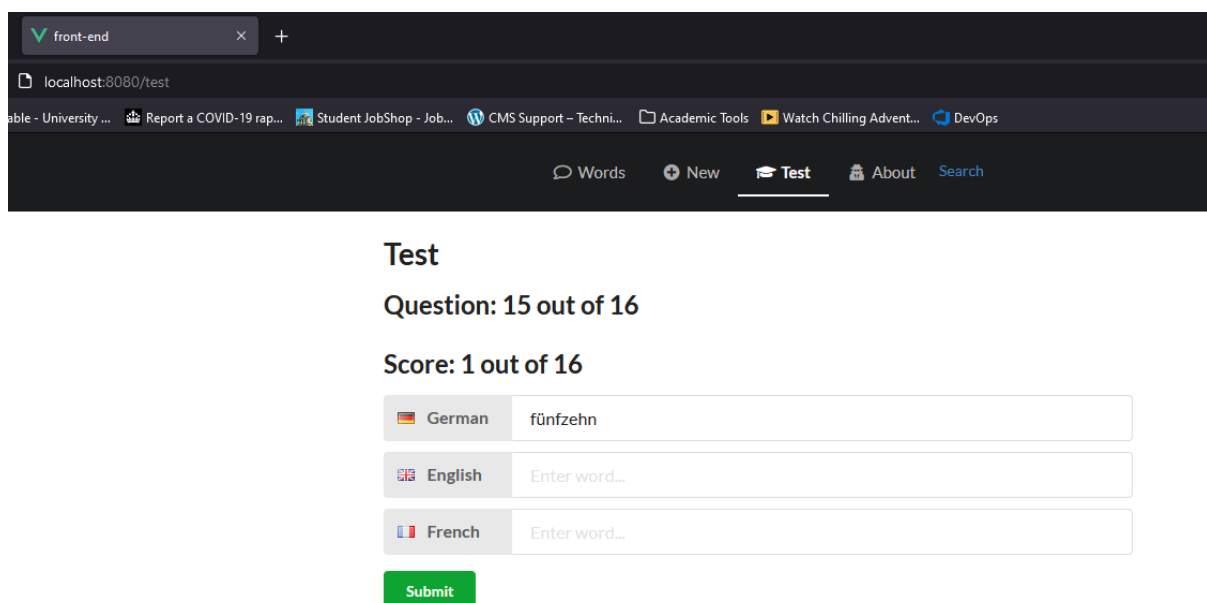


Figure 35 - Test view after 14 incorrect answers

Words New Test About Search

## Test

Question: 16 out of 16

Score: 1.5 out of 16

|         |               |
|---------|---------------|
| German  |               |
| English | Enter word... |
| French  | Enter word... |

Submit

You got the following words half correct: fünfzehn  
You got the following words wrong: drei, eins, zwei, sieben, vier, acht, sechs, neun, elf, zehn, zwölf, vierzehn, dreizehn, test1

Figure 36 - Completed test, one correct answer, one half correct, rest incorrect

Figures 34 to 36 demonstrate the test functionality of the system. The system iterates through all words in the database in a random order, each time a question is attempted it is removed from the array and the next question presented. Each question shows a word in German and the user must attempt to provide the correct English and French translations.

If the user gets the answer fully correct for all languages, they gain a single point. If they are correct with one language and not the other, they get half a point.

At the end of the test all incorrect words in German are displayed so the user may refer back to them for further study.